

Intermediate Macroeconomics: Lecture 10 (Ch.16)

ECON 311 – Intermediate Macroeconomics

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What are You Going to Learn (Ch. 16)

- The neoclassical consumption model
- Individuals choose the time path of their consumption to maximize utility
- Tow this model leads to a solution in which consumption is proportional to an individual's total wealth

Introduction to Consumption Model

- Consumption is the largest component of GDP (more than two-thirds)
- How do households decide what goods and services to consume?
 - In the Solow model, individuals spend a constant fraction of income (i.e., $C_t = (1 - \bar{s})Y_t$)
 - We should model how this $\bar{s}$ is formula, instead of taking it as exgenously given
- Heterogeneity in consumer behavior at the microeconomic level
 - The rich follow the permanent-income hypothesis
 - The poor typically have consumption that is quite sensitive to current income
- The consumption and saving are interlinked
 - We observe a decline in the personal saving rate and the rise in the debt-income ratio in recent decades

Neoclassical consumption model

- Individuals choose consumption at each moment in time
- Objective: maximize a lifetime utility function
 - Function depends on current and future consumption
- People recognize that income in the future may differ from income today
 - Such differences influence consumption today

The Neoclassical Consumption Model

- Two time periods: today and the future
 - People earn income and consume in both periods
- Key decision to make
 - How much to consume today versus how much to consume in the future
- The consumption model is based on two main elements:
 - An intertemporal budget constraint
 - inter: between/across
 - temporal: time
 - A utility function: a function that give us a measure of welfare

The Intertemporal Budget Constraint

- We start with two budget constraint, current period and future period
 - Each of them following that consumption equals income less saving
- The consumer's current-period budget constraint:

$$c = y - (f' - f)$$

where

c is the consumption

f is the amount of wealth at the start of current period

f' is the amount of wealth at the start of future period

y is the amount of income

- The $(f' - f)$ is the change in wealth = saving/borrowing

The Intertemporal Budget Constraint

- In the last period (future) there is nothing to save for, so all wealth and income are used for consumption
- In general, we use the prime symbol " ' " to denote the variable in future period
- The consumer's future-period budget constraint:

$$c' = y' + (1 + R)f',$$

where

R is the real interest rate

The Intertemporal Budget Constraint

- Arrange the future period budget constraint

$$f' = \frac{c' - y'}{(1 + R)}$$

- Then, sub it into the current period budget constraint

$$c = y - \left(\frac{c' - y'}{1 + R} - f \right)$$

- Put the consumption terms (both period) on the L.H.S. and the rest to the R.H.S

$$c + \frac{c'}{(1 + R)} = y + f + \frac{y'}{(1 + R)}$$

The Intertemporal Budget Constraint

- Human wealth is the present discounted value (PDV) of labor income

$$\text{Present value of consumption} = \text{Financial wealth} + \text{Human wealth}$$

- Consumption in a given year can be different from income.
- PDV of consumption must equal lifetime resources
- Students anticipate an increase in future income and smooth their consumption
 - By borrowing against future labor income, college students can consume more today
 - However, the present value of consumption must equal the present value of lifetime resources

- Utility depends on consumption today and in the future
- More consumption means higher utility

$$U = u(c) + \beta u(c')$$

where β is the discount factor or the weight placed on future consumption

- Diminishing Marginal Utility
 - Each additional unit of consumption raises utility by a smaller amount
- Usually, the $0 < \beta < 1$ so that future consumption is discounted – Today's consumption is valued more than future consumption

Flow Utility $u(c)$

- A consumption level of c delivers a flow of utility to the consumer of $u(c)$
- Utility rises when c increases, but the amount of the increase gets smaller and smaller, reflecting diminishing marginal utility

Choosing Consumption to Maximize Utility

- Maximize utility by choosing current period consumption and future period consumption, subject to a budget constraint

$$\max_{c, c'} U = u(c) + \beta u(c')$$

s.t.

$$c + \frac{c'}{(1+R)} = y + f + \frac{y'}{(1+R)} = \bar{X}$$

where $\bar{X}$ represents the total wealth

Choosing Consumption to Maximize Utility

- Rearrange the budget constraint:

$$c = \bar{X} - \frac{c'}{(1+R)}$$

- Sub it into the utility function, the maximization problem becomes

$$\max_{c'} U = u\left(\bar{X} - \frac{c'}{(1+R)}\right) + \beta u(c')$$

- First order conditions (take the derivative w.r.t. c' and set it equal to zero):

$$\frac{du\left(\bar{X} - \frac{c'}{(1+R)}\right)}{dc} \frac{-1}{1+R} + \beta \frac{du(c')}{dc'} = 0$$

The first term here we use the chain rule of calculus. Let $u'(c') = \frac{du(c')}{dc'}$, etc

$$\beta u'(c') = \frac{u'(c)}{1+R}$$

Choosing Consumption to Maximize Utility

- We want to make ourselves as happy as possible, but have a limited income
- Marginal utility of today's consumption: $u'(c)$
- Marginal utility of future consumption: $u'(c')$
- As seen from the equation from previous slides, agents can either
 - consume 1 unit today or
 - save that unit and consume $1 + R$ units in the future
- If individuals maximize utility, agents must be indifferent between consuming today or in the future

Choosing Consumption to Maximize Utility

- The equation that govern the trade-off between current and future consumption is called Euler equation:

$$\beta(1 + R)u'(c') = u'(c)$$

- MU of consuming 1 more unit today = discounted consumption of units in the future
 - LHS: Saving that same unit and consuming $1+R$ units in the future gives extra utility (discounted)
 - RHS: Consuming one unit today, the agent gets the MU of c today
- Must be indifferent between consuming today and tomorrow
- If the MU of c is greater than the utility from consuming the extra unit (plus interest) in the future,
 - the individual should reduce consumption in the future and raise it today, increasing overall lifetime utility
 - This reallocation should occur until the agent is just indifferent between consuming an extra unit today and consuming that unit plus interest in the future

Solving the Euler Equation

- Take $u(c) = \log(c)$ as an example
 - Recall that $\log(c) = \frac{1}{c}$
- This implies that $\beta(1 + R)u'(c') = u'(c)$ would become

$$\frac{\beta(1 + R)}{c'} = \frac{1}{c}$$

- Rearranging the Euler equation:

$$\frac{c'}{c} = \beta(1 + R)$$

- Left side is growth rate of consumption (plus 1)
 - Consumption is chosen such that the growth rate of consumption (c'/c) is a product of the discount parameter and the interest rate
- Agent takes R as given in partial equilibrium consumption problem, choosing the consumption growth rate

Solving the Euler Equation

$$\frac{c'}{c} = \beta(1 + R)$$

- Lower β : Impatient, consumption growth is lower
- Higher R : Patient, consumption growth is faster
- Suppose the economy is in steady state (i.e., $c'/c = 1$), then

$$\beta = \frac{1}{1 + R}$$

Solving the Model

- Rearrange the Euler equation:

$$c' = \beta(1 + R)c$$

- Plug it into the budget constraint

$$c + \frac{\beta(1 + R)c}{(1 + R)} = y + f + \frac{y'}{(1 + R)}$$

$$\Rightarrow c^* = \left[y + f + \frac{y'}{(1 + R)} \right] \frac{1}{1 + \beta} = \bar{X} \frac{1}{1 + \beta}$$

- Plus c^* into the Euler equation (alternative, you can plug it into the budget constraint)

$$c'^* = \left[y + f + \frac{y'}{(1 + R)} \right] \frac{\beta(1 + R)}{1 + \beta} = (1 + R)\bar{X} \frac{\beta}{1 + \beta}$$

The Effect of a Rise in R on Consumption

- Wealth Effect: A higher interest rate will reduce this present value, therefore reducing consumption in the case of log utility
 - Total wealth depends on the interest rate because of PDV of labor income (i.e., $\frac{y'}{1+R}$)
- Substitution effect: When interest rate increase, current consumption become more expensive (because saving will lead to even more future consumption), so consumers will reduce current consumption today
- Income effect: consumers are now richer—because their current saving leads to more income in the future—which makes them want to consume more today
- With log utility, the IE and SE offset each other
- Consumption could increase or decrease, depending on utility function

Lessons from The Neoclassical Model

- The permanent-income hypothesis is an implication of this model
 - Consumption depends on average income (PDV of income) rather than current income
- Recall overall wealth: $\bar{X} = f + y + \frac{y'}{1+R}$

Last two terms are PDV of income

- Consumption is proportional to a consumer's overall wealth

Lessons from The Neoclassical Model

- Permanent-income hypothesis implies that we want to smooth our consumption over time
- What happens if income increases?
 - We may spend some of the extra income, but will save some for the future
- Consumer do not like variability in consumption
- Marginal propensity to consume: portion of additional income that is spent for consumption
 - In the log example, it is equal to $\frac{1}{1+\beta}$ for c and $\frac{\beta}{1+\beta}$ for c'

The Desire to Smooth Consumption

- Suppose an individual could either consume c_1 today and c_2 in the future, or could consume the average of these two values in both periods
- Because of diminishing marginal utility, agents prefer to smooth consumption and take the average in both periods
- That is, lifetime utility is higher when consumer gets $u(\bar{c})$ in both periods, rather than $u(c_1) + u(c_2)$

Ricardian Equivalence

- Consumption depends on PDV of taxes and is invariant to the timing of taxes
- The government has a similar constraint that the lifetime government spending is also equal to lifetime tax revenue
- A tax cut today financed by higher future taxes will not affect consumption, as long as the lifetime government spending is the same

$$c + \frac{c'}{(1+R)} = y - t + f + \frac{y' - t'}{(1+R)} = \bar{X}$$

- Think of there is an increase in t with a decrease in t' of the same size (apply the interest rate)

$$c + \frac{c'}{(1+R)} = y - (t + \Delta t) + f + \frac{y' - (t' - (1+R)\Delta t)}{(1+R)}$$

- A change in the timing of taxes does not change wealth, so consumption doesn't change

Borrowing Constraints

- The model assumes that anyone can borrow or save at the interest rate R
- However, in reality, some people may not be able to borrow because of
 - poor financial wealth or no access to credit
 - weak economic conditions as a whole
- In either of these cases, the intertemporal budget constraint is no longer the correct constraint
- The constraint on consumption for individuals with no financial wealth and no access to credit can be rewritten as

$$c \leq y$$

- The inequality assumes people are constrained by the lack of borrowing opportunities to be no greater than a person's income

Borrowing Constraints

- If a consumer is already consuming less than their income, the no-borrowing constraint will be nonbinding
 - The individual is already saving, so not allowing him or her to borrow does not change anything
- If income is low and the borrowing constraint binds, then $c = y$
 - In most binding cases, it is undesired outcome
 - People cannot smooth their consumption even if they want to

Case Study: Consumption versus Expenditure

- Key prediction of model: Consumption should not change when an anticipated event occurs
 - Consumer wants to smooth out their consumption over time
- However, this doesn't always hold true
 - A retirement is usually not an unexpected event
 - Consumption should remain relatively unchanged when people retire
 - However, expenditures on consumption change quite markedly around this event, falling by around 17 percent
- A study by Aguiar and Hurst look at it differently
 - Use data set of food diaries for a large number of households
 - While expenditures on consumption do indeed decline sharply upon retirement, the food and calories that people actually consume show no such decline
 - Instead, households spend much more time shopping for food and preparing it themselves

Precautionary Saving

- Saving money to hedge against the possibility of a drop in income due to uncertain and unpredictable event
 - Consumers might save when income and wealth are temporarily low, meanwhile the basic permanent-income hypothesis would suggest borrowing
- The recent financial crisis provides an excellent example of precautionary saving
 - Saving rates rose sharply during the financial crisis, and precautionary motives are a logical part of the explanation

Next Lecture

- Next lecture will begin Ch.7